Abenezer Bekele

DATA ENGINEERING INTERN

Phone: (651) 399-6145
Email: AbenezerYBekele@Gmail.com
Address: Eagan, MN
Linkedin: linkedin.com/in/abenezer-bekele
GitHub: github.com/AbenezerYBekele

PROFESSIONAL SUMMARY

Detail-oriented Computer Science graduate with hands-on experience building ETL/ELT pipelines, designing relational databases, and developing data integration solutions. Proficient in Python and SQL for data transformation, quality assurance, and workflow automation. Experienced with AWS cloud services, Git-based version control, and Agile/Scrum development. Proven ability to collaborate cross-functionally and communicate technical findings to non-technical stakeholders. U.S. Naturalized Citizen. no sponsorship required

EDUCATION

University of Wisconsin – River Falls | River Falls, WI

- Bachelor of Science in Computer Science and Information Systems | December 2024
  Relevant Coursework: Database Systems, Data Science, Software Engineering, Object-Oriented Programming, Data Structure Algorithms

CERTIFICATIONS

- IBM data science professional certificate
- Google Data Analytics Professional Certificate
- AWS Cloud Practitioner Essentials (In Progress)

WORK EXPERIENCE

Development Intern | A Better Road Foundation

- Designed, developed, and deployed end-to-end web platform; analyzed datasets to support data-informed email and web engagement strategies.
- Built custom interactive elements using JavaScript (Velo), ensuring data integrity and responsive user experience across platforms.
- Integrated Google Analytics and implemented SEO best practices, providing leadership with data-driven insights on user engagement KPIs.
- Acted as sole technical liaison between stakeholders and IS team, gathering requirements and translating organizational goals into functional solutions.

TECHNICAL SKILLS

- **Languages:** Python (Pandas, NumPy, Scikit-learn), SQL (PostgreSQL, MS SQL Server, MySQL), JavaScript, TypeScript, Bash Scripting, R
- **ETL & Data Pipelines:** ETL/ELT Design, Data Cleaning, Data Integration, Data Transformation, Relational Database Design, Dimensional Modeling (Star/Snowflake)
- **Databases & Data Warehousing:** PostgreSQL, MS SQL Server, MySQL, MongoDB, DynamoDB, Google BigQuery (familiar), Snowflake (familiar), CTEs, Window Functions, Stored Procedures, Query Optimization
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, API Gateway), Docker, CI/CD Pipelines, Azure DevOps, Git/GitHub, Linux, Jira, Agile/Scrum
- **Data Quality & Governance:** Unit Testing (PyTest), Data Validation, Metadata Management, HIPAA Awareness, Data Privacy Compliance, NULL Filtering
- **BI & Visualization:** Power BI (DAX), Tableau, Matplotlib, Seaborn, Excel (Power Query, Pivot Tables, VBA/Macros), Jupyter Notebooks
- **Big Data & ML (Familiar):** Hadoop, Spark (familiar), Databricks (familiar), Airflow (familiar), Scikit-learn, Random Forest, Decision Trees

PROJECTS

Automated ETL & ML Pipeline: Titanic Survival & Drug Classification | GitHub

- **Data Quality & Wrangling:** Built a robust ETL pipeline for the Titanic and Drug-Classification datasets, handling missing values, feature scaling, and normalization to ensure model-ready data.
- **Predictive Analytics:** Developed and deployed machine learning models (Random Forest, Decision Trees) to predict outcomes, achieving high accuracy through hyperparameter tuning.
- **Unit Testing:** Applied source control (Git) and modular coding practices to ensure the pipeline was reusable and maintainable.

Relational Database Design & SQL Optimization (COVID-19) | GitHub

- **Advanced SQL:** Authored 12+ complex SQL queries using Window Functions, CTEs, and Self-Joins to transform raw global health data into structured analytical views.
- **Performance Tuning:** Optimized query execution by implementing strategic indexing and stored procedures, improving data retrieval speeds for regional summary reporting.
- **Data Governance:** Applied NULL filtering and data validation constraints to maintain high data integrity standards across all interfaces.

FinTracker: AI-Powered Financial API | GitHub

- **API Development:** Built a secure RESTful API using Node.js and PostgreSQL to facilitate data movement across systems.
- **GenAI Integration:** Leveraged Generative AI (Google Gemini Pro) to automate the extraction of metadata from unstructured images, optimizing the data ingestion workflow.
- **Containerization:** Utilized Docker for consistent deployment environments, ensuring scalability and security.

Public Interest Data Pipeline: Chicago Urban Analysis | GitHub

- **Complex Dataset Handling:** Engineered a data pipeline to process diverse urban datasets (socio-economic, education, and population) to identify service gaps in metropolitan areas.
- **Analysis & Modeling:** Utilized Python and Jupyter to clean and normalize "messy" public data, creating a clean core system for secondary analysis.
- **Visualization:** Developed automated reporting modules to communicate technical demographic trends to non-technical stakeholders.

Business Intelligence: Power BI Dashboard Portfolio | GitHub

- **End-to-End ETL:** Performed full ETL processes including data cleansing, validation, and transformation; optimized queries using indexing and stored procedures connected to live SQL databases.
- **KPI Reporting:** Built interactive KPI dashboards enabling real-time visibility into organizational performance metrics; documented data flows for ongoing cross-functional use.